Name: _____ General Physics

Unit 2: Vectors and Projectile Motion

Practice - Horizontal Launches (Sections 2.6–2.8)

Ignore air resistance; $a_y = -9.8\ \text{m/s}^2$. Launch point $x_i = 0$, ground $y = 0$ unless stated. Fill in the x and y columns, write the equation you start from, then plug in numbers with units (Desmos is fine for finding t). Problem 6 (★ **Challenge**) goes a step past the core skills.

1. Phoebe swats a pen off the edge of a desk. The pen leaves the desk moving horizontally at 1.6 m/s, and the desktop is 1.10 m above the floor.

x column	y column
$v_{ix} =$	$v_{iy} =$
$v_{fx} =$	$v_{fy} =$
$a_x =$	$a_y =$
$x_i =$	$y_i =$
$x_f =$	$y_f =$
$t =$ (same in both columns)	

Sketch (axes, launch point, landing point)

(a) How long is the pen in the air, and how far from the edge of the desk does it land?

(b) Find the speed and direction of the pen just before it hits the floor.

(c) Mr. Jantz claims that if Phoebe swats the pen twice as fast, it will (i) land twice as far from the desk and (ii) take half as much time to reach the floor. Decide whether each claim is right and explain using the x column and the y column.

2. A gardener holds a hose nozzle level, 1.20 m above the lawn. The water leaves the nozzle horizontally and lands 3.6 m away, measured along the ground.

x column	y column
$v_{ix} =$	$v_{iy} =$
$v_{fx} =$	$v_{fy} =$
$a_x =$	$a_y =$
$x_i =$	$y_i =$
$x_f =$	$y_f =$
$t =$ (same in both columns)	

Sketch (axes, launch point, landing point)

(a) How long does the water take to reach the lawn, and how fast is it moving when it leaves the nozzle?

2. *(continued)*

(b) At what angle below the horizontal does the water hit the lawn?

(c) **Predict.** The gardener crouches and holds the nozzle level 0.30 m above the lawn, one fourth as high, with the water leaving at the same speed. How far away does the water land now? Explain without a new calculation.

3. A rescue plane flies level at 55.0 m/s and releases a supply crate. The crate lands 220 m ahead of the point where it was released, measured horizontally.

x column	y column
$v_{ix} =$	$v_{iy} =$
$v_{fx} =$	$v_{fy} =$
$a_x =$	$a_y =$
$x_i =$	$y_i =$
$x_f =$	$y_f =$
$t =$ (same in both columns)	

Sketch (axes, launch point, landing point)

(a) How long is the crate in the air, and how high was the plane when it released the crate?

(b) How high above the ground is the crate 2.00 s after release? Has it fallen half of its height in half of its time? Explain why or why not.

(c) Where is the plane, relative to the crate, when the crate hits the ground? Explain.

4. Mr. Lehman rides a motorcycle horizontally off the roof of a 12.0 m tall building toward a 7.0 m tall building next door. The gap between the buildings is 9.0 m wide.

x column	y column
$v_{ix} =$	$v_{iy} =$
$v_{fx} =$	$v_{fy} =$
$a_x =$	$a_y =$
$x_i =$	$y_i =$
$x_f =$	$y_f =$
$t =$ (same in both columns)	

Sketch (axes, launch point, landing point)

(a) How long does Mr. Lehman fall before he reaches the height of the lower roof, and what is the smallest speed he can leave the roof with and still make it across the gap?

(b) He actually leaves the roof at 12.0 m/s. How far past the edge of the lower roof does he land?

(c) **Predict.** If the lower roof were 2.5 m below the upper roof instead of 5.0 m, would his minimum speed be twice as large, half as large, about 1.4 times as large, or the same? Explain without a new calculation.

5. Marble A rolls off a lab table at 2.0 m/s. Marble B rolls off a shelf four times as high as the table at 1.0 m/s. For each comparison, write *greater than*, *less than*, or *equal to*, then explain. Name the column you use.

Marble A

x column	y column
$v_{ix} =$	$v_{iy} =$
$v_{fx} =$	$v_{fy} =$
$a_x =$	$a_y =$
$x_i =$	$y_i =$
$x_f =$	$y_f =$
$t =$ (same in both columns)	

Marble B

x column	y column
$v_{ix} =$	$v_{iy} =$
$v_{fx} =$	$v_{fy} =$
$a_x =$	$a_y =$
$x_i =$	$y_i =$
$x_f =$	$y_f =$
$t =$ (same in both columns)	

(a) The time Marble A is in the air is _____ the time Marble B is in the air.

5. *(continued)*

(b) Marble A's horizontal final position x_f is _____ Marble B's.

(c) The size of Marble A's vertical velocity just before it lands is _____ Marble B's.

(d) The angle below the horizontal at which Marble A lands is _____ Marble B's.

6. ★ **Challenge.** Ivy jumps horizontally off the edge of Mr. Jantz's porch. She lands on the grass moving at 4.50 m/s at 35.0° below the horizontal.

x column	y column
$v_{ix} =$	$v_{iy} =$
$v_{fx} =$	$v_{fy} =$
$a_x =$	$a_y =$
$x_i =$	$y_i =$
$x_f =$	$y_f =$
$t =$ (same in both columns)	

Sketch (axes, launch point, landing point)

(a) Find Ivy's horizontal and vertical velocity components just before she lands.

(b) Is Ivy's speed when she leaves the porch greater than, less than, or equal to 4.50 m/s? Explain.

(c) How long is Ivy in the air, and how high is the porch?

(d) How far from the edge of the porch does Ivy land?